

Quiz (Habanero)

Q1. What is the expansion of $(x + 5)^2$?

- (A) $x^2 + 10x + 25$
 (B) $x^2 - 10x + 25$
 (C) $x^2 + 5x + 10$
 (D) $x^2 - 5x - 10$
 (E) $x^2 + 25$

Q2. If $y = 3x^2 + 12x + 12$, is this a perfect square trinomial?

- (A) Yes
 (B) No
 (C) Maybe
 (D) Insufficient information
 (E) Only for certain values of x

Q3. What is the factorization of $x^2 - 14x + 49$?

- (A) $(x - 7)^2$
 (B) $(x + 7)^2$
 (C) $(x - 7)(x + 7)$
 (D) $x(x - 14) + 49$
 (E) $x(x + 14) - 49$

Q4. The temperature of a city can be modeled by $T(x) = x^2 - 4x + 4$, where x is the number of hours after sunrise. What is the factored form of $T(x)$?

- (A) $(x - 2)^2$
 (B) $(x + 2)^2$
 (C) $x(x - 4)$
 (D) $(x - 4)(x + 1)$
 (E) $(x + 2)(x - 2)$

Q5. A weather balloons' altitude can be represented as $h =$ $t^2 - 6t + 8$. Is the expression h a perfect square trinomial?

- (A) Yes
 (B) No
 (C) For $t > 0$
 (D) Only if t is a real number
 (E) Not enough information

Q6. The area of a rectangular garden is $A = x^2 + 10x + 25$. What is the length of the sides of this garden?

- (A) 5m and 5m
 (B) 10m and 10m
 (C) 5m and 10m
 (D) x m and $(x + 5)$ m
 (E) $(x + 5)$ m and $(x + 5)$ m

Q7. Water level of a river can be represented as $W(x) = x^2 - 8x + 16$, where x is the distance in kilometers from the dam. What is $W(4)$?

- (A) 0
 (B) 4
 (C) 8
 (D) 16
 (E) 32

Q8. In an ecosystem, population of a certain species is given by $P(x) = x^2 + 2x + 1$. Factorize the population equation.

- (A) $(x + 1)^2$
 (B) $(x - 1)^2$
 (C) $(x + 2)(x + 1)$
 (D) $(x - 2)(x - 1)$
 (E) $(x + 1)(x - 1)$

Open Ended Questions (FRQ)

Q9. The cost to remove a certain amount of pollution from a lake can be modeled as $C(x) = x^2 - 12x + 36$, where x is the amount of pollution in tons. If $x = 6$ tons of pollution has been removed, what is $C(6)$ and what does it represent in the context of the problem. Show your work and explain your answer.

Q10. A researcher models the effect of temperature T (in $^{\circ}\text{C}$) on the growth rate G (in

Chef's Tips

- Emphasize the importance of checking the work.
- Use real-life scenarios whenever possible to demonstrate practical applications.
- Provide step-by-step solutions for complex problems.